

Sensor and Systems

Sensor fusion for distributed systems

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Electronic Systems

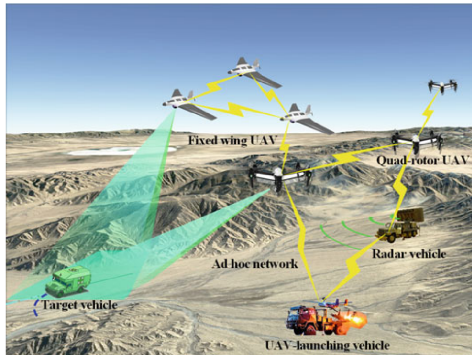
8th semester

- ▶ State estimation in distributed systems
- ▶ Example of distributed sensor fusion
- ▶ Non-concurrent sampling
- ▶ Estimation in multi-rate sensor networks



From www.ignitec.com:

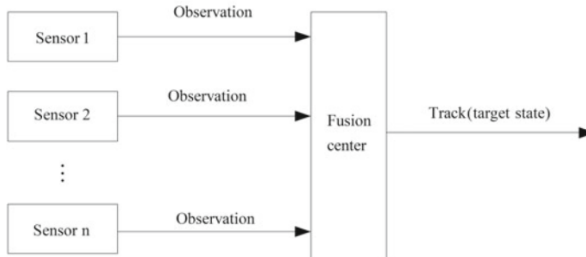
- ▶ “Body sensor networks integrate the technology of sensing, intelligent information processing, pervasive computing, and communication
- ▶ Body sensor devices process information and provide a near-accurate view of the physiological, behavioural, health, or emotional state of the person wearing it
- ▶ Body sensors as measurement devices. For example, heart rate, oxygen saturation, motion, blood pressure, arterial pressure, electrical or sweat gland activity [...]”



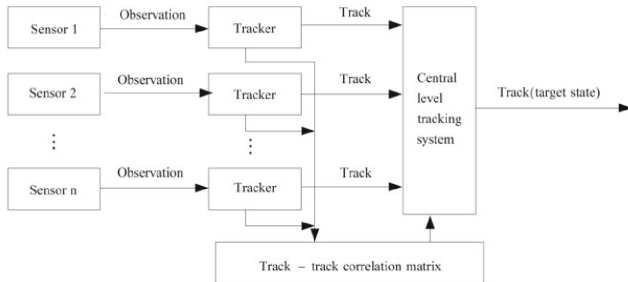
- ▶ Several mobile units cooperate to achieve a common goal, e.g., tracking of some target;
- ▶ However, since the individual units move around, communication between units becomes a dynamic problem.

Distributed sensing

Centralized sensor fusion



- ▶ Individual agents acquire sensor data;
- ▶ The 'raw' sensor data is transmitted to a centralized station where the fusion takes place (potentially along with noise covariance estimates).
- ▶ Same as centralized sensor fusion, except for the presence of a network.



- Each agent accumulates data from its own sensor(s).
- Each agent provides its own estimate of the target state.
- The target estimates can then be communicated to a central processor that fuses the estimates.

- Each agent accumulates data from its own sensor(s).
Agent $i, i = 1, 2, \dots, p_s$ assumes that the target dynamics follows the same LTI model:

$$\begin{aligned}x_{k+1} &= \Phi x_k + \Gamma u_k + w_k, \quad w \in \mathcal{N}(0, Q) \\ y_{i,k} &= H_i x_k + \nu_{i,k}, \quad \nu \in \mathcal{N}(0, R)\end{aligned}$$

- Each agent provides its own estimate of the target state – for instance using a local Kalman filter:

$$\begin{aligned}K_i &= P_{i,k|k-1} H_i^\top (H_i P_{i,k|k-1} H_i^\top + R_i) \\ P_{i,k|k} &= (I - K_i H_i) P_{i,k|k-1} (I - K_i H_i)^\top + K_i R_i K_i^\top \\ \hat{x}_{i,k+1} &= \Phi \hat{x}_{i,k} + \Gamma u_k + \Phi K_i (y_{i,k} - H_i \hat{x}_{i,k}) \\ P_{i,k+1|k} &= \Phi P_{i,k|k} \Phi^\top + Q\end{aligned}$$

- The target estimates can then be communicated to a central processor that fuses the estimates.

$$\hat{x}_k = \psi(\hat{x}_{1,k}, P_{1,k|k}, \hat{x}_{2,k}, P_{2,k|k}, \dots, \hat{x}_{p_s,k}, P_{p_s,k|k})$$

Example – Section 3.5

Very naive fusion



► Process

$$x_{k+1} = \begin{bmatrix} 0.95 & 0.95 \\ 0 & 0.95 \end{bmatrix} x_k + w_k, \quad w \in \mathcal{N} \left(0, \begin{bmatrix} \frac{0.01}{3} & \frac{0.01}{2} \\ \frac{0.01}{2} & 0.01 \end{bmatrix} \right)$$

$$y_{1,k} = \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + \nu_{1,k}$$

$$y_{2,k} = \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + \nu_{2,k}$$

$$y_{3,k} = \begin{bmatrix} 0 & 1 \end{bmatrix} x_k + \nu_{3,k}$$

► Correlated measurement noise:

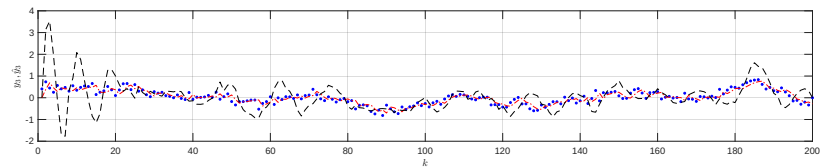
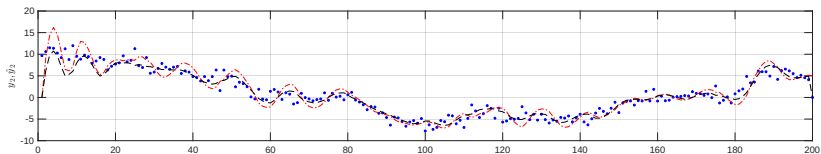
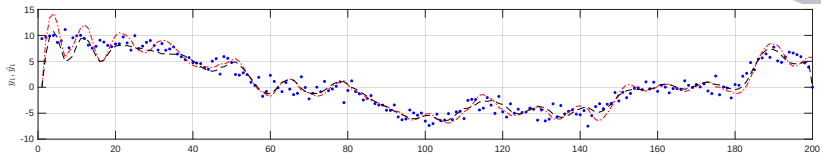
$$\nu \in \mathcal{N}(0, R) \quad \text{where} \quad R = \begin{bmatrix} 1 & 0.5 & 0.005 \\ 0.5 & 1 & 0.005 \\ 0.005 & 0.005 & 0.01 \end{bmatrix}$$

► Naive fusion:

$$\hat{x}_k = \frac{1}{p_s} \sum_{i=1}^{p_s} \hat{x}_{i,k|k}$$

Fusion performance – very naive approach

Dots: measurements; dash-dot: Local estimate, dashed: Fused



- ▶ Each state estimate is most likely not equally trustworthy.
- ▶ This is – to the best available knowledge – expressed through the covariance estimates $P_{i,k|k}$.
- ▶ Slightly less naive fusion:

$$\hat{x}_k = \hat{P}_{k|k} \sum_{i=1}^{p_s} P_{i,k|k}^{-1} \hat{x}_{i,k|k}$$

where $\hat{P}_{k|k} = \sum_{i=1}^{p_s} P_{i,k|k}$.

- ▶ Optimal(?) fusion according to Yan et. al.¹:

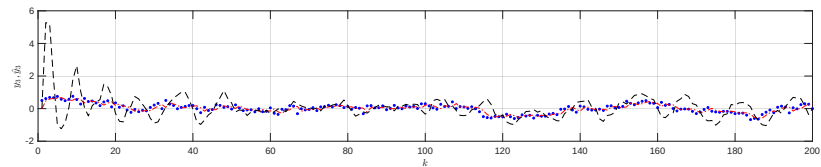
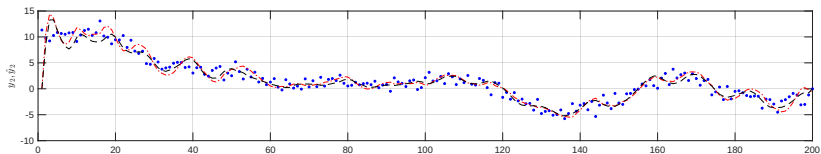
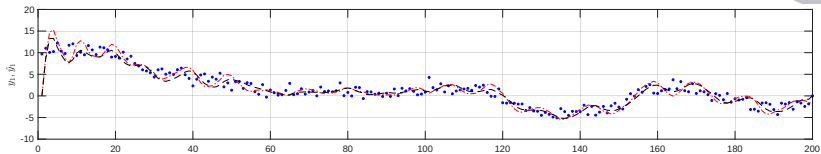
$$P_{i,k|k}^{-1} = P_{i,k|k-1}^{-1} + \sum_{i=1}^{p_s} (P_{i,k|k}^{-1} - P_{i,k|k-1}^{-1})$$

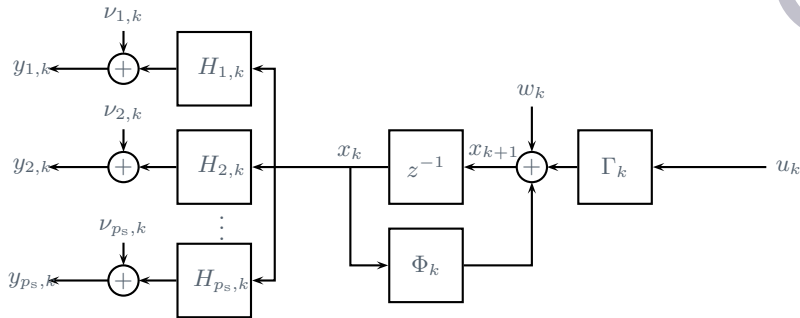
$$P_{i,k|k}^{-1} \hat{x}_{k|k-1} = P_{i,k|k-1}^{-1} \hat{x}_{k|k-1} + \sum_{i=1}^{p_s} (P_{i,k|k}^{-1} \hat{x}_{i,k|k} - P_{i,k|k-1}^{-1} \hat{x}_{i,k|k-1})$$

¹Note that the authors use back-in-time filter implementation, which can cause some implementation issues...

Fusion performance – slightly less naive approach

Dots: measurements; dash-dot: Local estimate, dashed: Fused





- In the discrete-time formulation of sensor fusion, measurements from all sensors are generically assumed to be available at every sample instance:

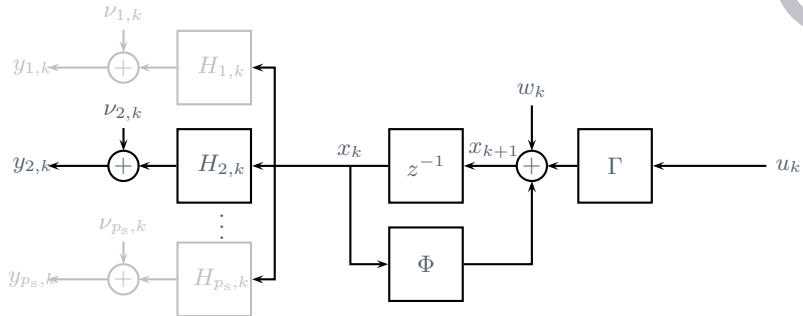
$$x_{k+1} = \Phi_k x_k + \Gamma_k u_k + w_k$$

$$y_{1,k} = H_{1,k} x_k + \nu_{1,k}$$

$$y_{2,k} = H_{2,k} x_k + \nu_{2,k}$$

$$\vdots$$

$$y_{p_s,k} = H_{p_s,k} x_k + \nu_{p_s,k}$$



- What to do if sensor measurements do not occur/arrive at the fusion center at the same time?

$$x_{k+1} = \Phi x_k + \Gamma u_k + w_k$$

$$y_{1,k} = H_{1,k} x_k + \nu_{1,k}$$

$$y_{2,k} = H_{2,k} x_k + \nu_{2,k}$$

$$\vdots$$

$$y_{p_s,k} = H_{p_s,k} x_k + \nu_{p_s,k}$$



$$y_k = H_k x_k + \nu_k$$

$$y_k = \begin{bmatrix} 0 \\ y_{j,k} \\ 0 \end{bmatrix}, \quad H_k = \begin{bmatrix} 0 \\ H_{j,k} \\ 0 \end{bmatrix}, \quad \nu_k = \begin{bmatrix} 0 \\ \nu_{j,k} \\ 0 \end{bmatrix}.$$

- Initial conditions: $x_0 \in \mathcal{N}(\hat{x}_{0|-1}, P_{0|-1})$.
- Measurement update after receiving y_k and u_k :

$$H_k = \text{select}(y_k)$$

$$\hat{y}_{k|k-1} = H_k \hat{x}_{k|k-1}$$

$$\tilde{y}_{k|k-1} = y_k - \hat{y}_{k|k-1}$$

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1}$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1}$$

$$P_{k|k} = (I - K_k H_k)^{-1} P_{k|k-1} (I - K_k H_k)^\top + K_k R_k K_k^\top.$$

- Time update from k to $k+1$:

$$\hat{x}_{k+1|k} = \Phi \hat{x}_{k|k} + \Gamma u_k$$

$$P_{k+1|k} = \Phi^\top P_k \Phi + Q$$

- Suppose $H_k^\top = \begin{bmatrix} 0 & H_j^\top & 0 \end{bmatrix}$; then

$$P_{k|k-1} H_k^\top = \begin{bmatrix} P_{11} & P_{1j} & P_{13} \\ P_{1j}^\top & P_{jj} & P_{j3} \\ P_{13}^\top & P_{j3}^\top & P_{33} \end{bmatrix} \begin{bmatrix} 0 & H_j^\top & 0 \end{bmatrix} = \begin{bmatrix} 0 & P_{k|k-1} H_j^\top & 0 \end{bmatrix}$$

$$H_k P_{k|k-1} H_k^\top = \begin{bmatrix} 0 \\ H_j \\ 0 \end{bmatrix} \begin{bmatrix} 0 & P_{k|k-1} H_j^\top & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & H_j P_{k|k-1} H_j^\top & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

- This yields a Kalman gain of the form

$$\begin{aligned} K_k &= P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1} \\ &= \begin{bmatrix} 0 & P_{k|k-1} H_j^\top & 0 \end{bmatrix} \begin{bmatrix} R_{11} & R_{1j} & R_{13} \\ R_{1j}^\top & R_{jj} + H_j P_{k|k-1} H_j^\top & R_{j3} \\ R_{13}^\top & R_{j3}^\top & R_{33} \end{bmatrix}^{-1} \\ &= \begin{bmatrix} \star & K_j & \star \end{bmatrix} \end{aligned}$$

► State:

$$\begin{aligned}\hat{x}_{k|k} &= \hat{x}_{k|k-1} + K_k \tilde{y}_{k|k-1} \\ &= \hat{x}_{k|k-1} + \begin{bmatrix} \star & K_j & \star \end{bmatrix} \begin{bmatrix} 0 \\ \tilde{y}_{j,k} \\ 0 \end{bmatrix} \\ &= \hat{x}_{k|k-1} + K_j \tilde{y}_{j,k}\end{aligned}$$

► Note: the covariance estimate $P_{k|k}$ is a bit more ‘muddy.’

Example

Centralized fusion, alternating sampling

► Process:

$$x_{k+1} = \begin{bmatrix} 0.95 & 0.95 \\ 0 & 0.95 \end{bmatrix} x_k + w_k, \quad w \in \mathcal{N}\left(0, \begin{bmatrix} \frac{0.01}{3} & \frac{0.01}{2} \\ \frac{0.01}{2} & 0.01 \end{bmatrix}\right)$$

$$y_{1,k} = \begin{bmatrix} 1 & 0 \end{bmatrix} x_k + \nu_{1,k}$$

$$y_{2,k} = \begin{bmatrix} 0 & 1 \end{bmatrix} x_k + \nu_{2,k}$$

► Correlated measurement noise:

$$\nu \in \mathcal{N}(0, R) \quad \text{where} \quad R = \begin{bmatrix} 1 & 0.005 \\ 0.005 & 0.01 \end{bmatrix}$$

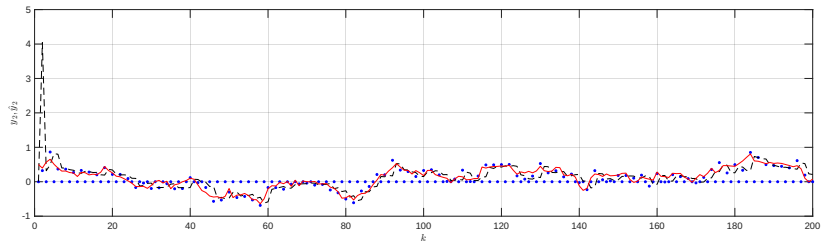
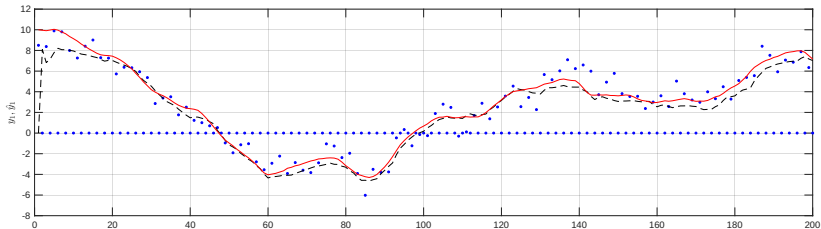
► Measurements:

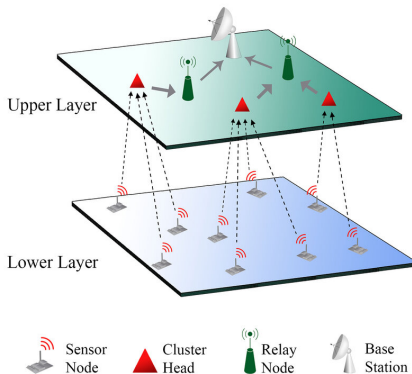
$$\text{If } k \text{ is odd: } y_k = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} x_k + \begin{bmatrix} \nu_{1,k} \\ 0 \end{bmatrix}$$

$$\text{If } k \text{ is even: } y_k = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} x_k + \begin{bmatrix} 0 \\ \nu_{2,k} \end{bmatrix}$$

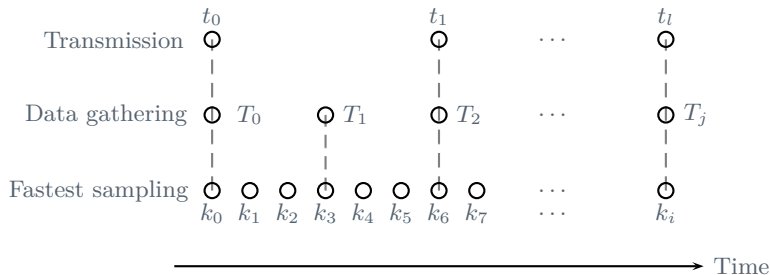
Fusion performance – alternating sampling

Full line: states, dots: measurements; dashed: fused estimates





- Often, sensor networks are *hierarchical* as shown above.
- Data is gathered by individual sensors and transmitted to local data hubs, which relay the gathered data to a base station.
- The data hubs provide local resilience, but also introduce delays.
- Other factors, such as retransmissions and other forms of protocol overhead, can also contribute to variability in arrival times at the base station.



- Fastest sampling occurs at intervals $h_e = k_{i+1} - k_i$.
- Data gathering occurs at intervals $h_m = T_{j+1} - T_j$.
- Transmission occurs at $h_p = t_{l+1} - t_l$.
- Multirate system: $\exists a, b \in \mathbb{Z}_+ : h_m = bh_e = abh_p$.

- We assume a fast LTI model:

$$\begin{aligned}x_{k_i+1} &= \Phi x_{k_i} + \Gamma w_{k_i}, & w &\in \mathcal{N}(0, Q) \\y_{s,k_i} &= H_s x_{k_i} + \nu_{s,k_i}, & \nu &\in \mathcal{N}(0, R)\end{aligned}$$

where s denotes sensor number.

- Data gathering occurs at times T_j , corresponding to the a -step model

$$\begin{aligned}x_{T_{j+1}} &= \Phi^a x_{T_j} + \begin{bmatrix} \Phi^{a-1}\Gamma & \Phi^{a-2}\Gamma & \dots & \Gamma \end{bmatrix} \begin{bmatrix} w_{T_i} \\ w_{T_i+h_p} \\ \vdots \\ w_{T_i+(a-1)h_p} \end{bmatrix} \\ &= \Phi_e x_{T_j} + \Gamma_e w_{e,T_j}\end{aligned}$$

- Data is received at times t_l , corresponding to the b -step model

$$\begin{aligned}x_{t_{l+1}} &= \Phi_e^b x_{t_l} + \begin{bmatrix} \Phi_e^{b-1}\Gamma_e & \Phi_e^{b-2}\Gamma_e & \dots & \Gamma_e \end{bmatrix} \begin{bmatrix} w_{e,t_l} \\ w_{e,t_l+h_e} \\ \vdots \\ w_{e,T_i+(b-1)h_e} \end{bmatrix} \\ &= \Phi_m x_{t_l} + \Gamma_m w_{m,t_l}.\end{aligned}$$

- Stacking delayed copies of the latter model then yields the augmented LTI model:

$$\begin{bmatrix} x_{t_l+1} \\ x_{t_l+1-h_e} \\ \vdots \\ x_{t_l+1-(b-1)h_e} \end{bmatrix} = \begin{bmatrix} \Phi_e^b & 0 & \cdots & 0 \\ \Phi_e^{b-1} & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_e & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_{t_l} \\ x_{t_l-h_e} \\ \vdots \\ x_{t_l-(b-1)h_e} \end{bmatrix} \\
 + \begin{bmatrix} \Phi_e^{b-1}\Gamma_e & \Phi_e^{b-2}\Gamma_e & \cdots & \Gamma_e \\ \Phi_e^{b-2}\Gamma_e & \Phi_e^{b-3}\Gamma_e & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \Gamma_e & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} w_{e,t_l} \\ w_{e,t_l+h_e} \\ \vdots \\ w_{e,T_i+(b-1)h_e} \end{bmatrix}$$

which can be used for sensor fusion at the top layer.

- The equations for the associated Kalman filter are symbolically familiar territory, but of course somewhat higher-dimensional than what we have seen until now.

- ▶ Network topology
- ▶ Network *consensus*
- ▶ Uncertainty (bit errors, delays etc.) induced by transmission across network
- ▶ Non-white noise, imperfect models
- ▶ Loss of connection, in particular in case of wireless sensor networks
- ▶ ...

- ▶ Sensor networks consist of sensor devices that are distributed in some way, and able to communicate their sensor measurement to a so-called *fusion center*.
- ▶ It is possible to estimate states locally, but one still has to fuse the individual state estimates.
- ▶ If samples do not arrive at the same time, one potentially has a problem.
- ▶ Fortunately, the Kalman filter handles this in a straightforward manner.
- ▶ Multi-rate sensor networks may be handled by multi-step prediction filters and *lifting*.